

■ Problem Statement

Creating professional explainer videos is expensive, slow, and cloud-dependent. Tools like HeyGen or Synthesia cost █5,000–█50,000 per video and require uploading private content to the internet. For educators, coaches, and small businesses in low-bandwidth environments — this is a hard wall.

Meanwhile, modern AI-capable Snapdragon X Elite laptops possess **enough on-device NPU compute** to run language models at 22+ tokens/second — yet no tool exists that turns this power into ready-to-share video content, fully offline.

■ Solution — SnapVid

SnapVid is an **offline, agentic AI explainer-video maker** that runs entirely on a Snapdragon X Elite laptop — no cloud, no subscriptions, no internet. It converts a single plain-text prompt into a **polished, narrated, animated MP4 video in under 60 seconds** using a two-agent pipeline.

Three deployment modes — all without leaving the device:

- **Browser UI** — Dark-mode studio dashboard with live SSE agent logs, settings modal, history, and built-in video player.
- **CLI** — `python run.py "your topic"` for headless pipelines and batch generation.
- **Python API** — Importable module for custom applications, bots, and agent loops.

■ Technical Architecture

SnapVid is built with **zero cloud runtime dependencies** using only Python, Pillow, NumPy, and bundled FFmpeg:

Layer	Technology	Capability
Scene Planner Agent	Qwen3-4B · GenieX NPU (w4a16)	JSON scene script + self-healing repair loop (3 retries)
Quality Critic Agent	Second LLM pass · same model	Autonomous narration & pacing review
Frame Renderer	Pillow + NumPy · 1280×720 @ 30 FPS	10 templates: title, bullets, bar chart, counter, kinetic, code, split, quote, icon list, lower-third
Scene Transitions	Pure-Python compositor	Fade, slide, zoom-in, cross-dissolve between scenes
TTS Engine	Kokoro 82M → Edge TTS → SAPI	Auto-select offline neural voice; word-level caption sync
Video Mux	FFmpeg (imageio-ffmpeg bundled)	H.264 / AAC MP4 — no system ffmpeg install required
Real-time Events	Server-Sent Events (SSE)	Live token, validation, render & TTS progress streamed to UI
Test Suite	pytest · 74 tests	Schema, LLM, renderer, TTS, server, transitions, history, NPU

■ Snapdragon NPU Performance Benchmarks

Verified on **Snapdragon X Elite CRD (Hexagon NPU)** via Qualcomm AI Hub cloud profiling with GenieX w4a16 quantization:

Component	Hardware	Latency	Throughput
Qwen3-4B Scene Planner	Hexagon NPU (GenieX w4a16)	~2.1 s / script	~22.4 tok/s
Kokoro TTS (per scene)	CPU ARM64	~1.8 s / scene	Real-time ×1.4
Pillow Frame Renderer	CPU ARM64	~0.6 s / scene	30 FPS native
Full Pipeline (60 s video)	Snapdragon X Elite	~30 s total	2× Real-Time

■ Key Differentiators vs. Existing Solutions

Feature	SnapVid	HeyGen / Synthesia	Manim / Remotion
Runs offline	■ Yes — full NPU	■ Cloud required	■■ Partial
Cost	Free · MIT Open Source	\$24–\$119/month	Free but expert setup
Privacy	■ Data never leaves device	■ Uploads your content	■ Local
Setup time	pip install + 1 command	Account + subscription	Hours (LaTeX/Node deps)
Video in 60 s	■ ~30 s on Snapdragon	■■ Minutes (cloud queue)	■ Hours (render compile)
Snapdragon NPU	■ Hexagon NPU via GenieX	■ No	■ No

■ Target Users & Use Cases

- **School & Coaching Institute Teachers** — Generate curriculum explainer videos for offline classrooms.
- **YouTube Educators** — Rapid STEM and tutorial video drafts without a studio.
- **Corporate L&D; Teams** — Internal training modules with zero data exposure.
- **Small Business Owners** — Product explainers in local language, offline, at zero cost.
- **Developers & Data Scientists** — Automated documentation and concept visualization pipelines.

■ Technology Stack

Category	Technology
Language / Runtime	Python 3.11+
LLM Inference (NPU)	Qualcomm GenieX · Hexagon NPU · w4a16 quantization · Qwen3-4B-Instruct
LLM Inference (CPU/Cloud)	Ollama · llama.cpp · OpenAI · Groq · NVIDIA NIM · Google Gemini
Frame Rendering	Pillow ≥10.0 · NumPy ≥1.26 · Custom easing engine
Video Encoding	FFmpeg (bundled via imageio-ffmpeg) · H.264 + AAC · MP4
Text-to-Speech	Kokoro 82M (neural, offline) · Edge TTS · Windows SAPI
Schema Validation	Pydantic v2 · JSON repair loop (3 retries)
Web Server / API	FastAPI · SSE streaming · CORS · REST endpoints
Frontend	Vanilla HTML5 / CSS3 / JS · Aurora UI · Plus Jakarta Sans · JetBrains Mono

Category	Technology
Testing	pytest · 74 tests · 100% passing
License	MIT Open Source

■ Innovation Highlights

- **Self-Healing Agentic Pipeline** — Invalid JSON from the LLM triggers an automatic repair prompt loop (up to 3 retries) before failing.
- **Two-Agent Architecture** — Scene Planner agent + independent Quality Critic agent reviewing pacing, clarity, and narration.
- **Audio-First Rendering** — Narration is synthesised first, scene durations are reconciled to audio, then frames are rendered — guaranteeing perfect A/V sync.
- **Word-Level Caption Overlay** — On-screen captions are timed to individual word offsets within each audio chunk, providing karaoke-style display.
- **Animated Scene Transitions** — Smooth fade, slide, zoom-in, and cross-dissolve transitions composited in pure Python between scenes.
- **Configurable at Runtime** — LLM provider, model, API key, voice engine, and video format all switchable live from the web UI — no code changes required.

■ Roadmap

- **Multilingual TTS** — Hindi & regional language voice via Meta MMS-TTS.
- **On-Device Captions via Whisper NPU** — Animated karaoke subtitles transcribed locally on Snapdragon NPU.
- **GIF / WebM / Vertical 9:16 Export** — Mobile-first shorts and social media clips.
- **Slide Import** — PPTX → video pipeline using existing template system.